

House Price Prediction

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Objective

Build a regression model that predicts house prices based on property features (area, bedrooms, bathrooms, amenities) and identify which features most strongly influence property value.

Dataset

The analysis uses the Housing Prices Dataset from Kaggle (545 properties, 13 features). Features include numerical (area, bedrooms, bathrooms, stories, parking) and categorical (mainroad, guestroom, basement, hotwaterheating, airconditioning, prefarea, furnishingstatus). No missing values were found.

Methodology

- Data Cleaning: Converted binary categoricals (yes/no to 1/0), applied one-hot encoding to furnishing status, verified no missing values or duplicates.
- Feature Engineering: Created total_amenities (count of premium features) and luxury_score (composite quality indicator combining bathrooms, stories, parking, amenities).
- Model Training: Trained three regression models on an 80/20 train-test split - Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor.
- Evaluation: Used MAE, RMSE, R-squared, and 5-fold cross-validation.

Key Results

Model	MAE	RMSE	R2 Score
Linear Regression	\$970,043	\$1,324,507	0.6529
Random Forest	\$918,425	\$1,333,773	0.6481
Gradient Boosting	\$953,398	\$1,352,140	0.6383

Best Model: Linear Regression with R2 = 0.6529 (explains ~65% of price variation)

Key Insights

- Area is the strongest individual predictor - square footage (importance: 0.227) confirms size drives price.
- Luxury score (engineered feature) dominates at 0.555 importance - proving feature engineering adds real value.
- Bathrooms matter more than bedrooms - buyers value luxury amenities over room count.
- Furnishing boosts value - unfurnished homes sell for significantly less.
- Hot water heating is irrelevant (correlation: 0.09), while air conditioning strongly impacts price (0.45).

Business Recommendation

- Prioritize area and bathrooms when evaluating or developing properties.

- Stage/furnish homes before listing - the price premium justifies the cost.
- Install air conditioning as a cost-effective value-add.
- Don't over-invest in hot water heating - it doesn't meaningfully impact selling price.

Tools Used

Python 3.x with Pandas, Scikit-learn, Matplotlib, Seaborn | Jupyter Notebook for interactive analysis | 3 Regression Models with 5-fold cross-validation | 6 Professional Visualizations saved as PNGs

This project demonstrates data loading, cleaning, feature engineering, model comparison, and business insight generation.